A PRACTICAL GUIDE FOR STUDENTS

Choose a starting point from the problem

No architecture is guaranteed to be best. Use the task and data to narrow candidates, then evaluate them.

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IMAGE / GRID

One label per image

Use classification for one label per image. For object locations or pixel masks, use detection or segmentation.

Output: class, boxes, or mask

TEXT / SEQUENCE

Order matters

Start from the output: classify, extract entities, translate, summarize, or generate. Choose a checkpoint for that task.

Output: label, spans, or text

TABULAR

Rows of features

Start with a simple baseline. Compare linear or tree-based methods before assuming a neural network is needed.

Output: class or number

TIME SERIES / AUDIO

Signals over time

Define the target and time horizon. Split by time when the goal is to predict future data.

Output: class, value, or sequence

A reliable selection loop

1 **Define the goal** — 2 **Check task + data fit** — 3 **Shortlist a baseline** — 4 **Compare on held-out data**

TOOLS AND REFERENCES

Use the tool that fits the work

Frameworks build and run models; an API simplifies the code; a hub helps you find models to reuse.

FRAMEWORKS / BUILD, TRAIN & RUN

TENSORFLOW

FRAMEWORK

Training-to-deployment ecosystem

Choose for end-to-end ML workflows, including training, production pipelines, and deployment. Keras is often the simplest entry point.

TensorFlow: guides and tutorials

PYTORCH

FRAMEWORK

Flexible, Python-first framework

Choose for custom models, hands-on training loops, and research-oriented experimentation. TorchVision adds pretrained vision models.

PyTorch: transfer learning tutorial

COMPLEMENTARY LAYERS / SIMPLIFY & REUSE

KERAS

HIGH-LEVEL API

Concise model-building interface

Choose for a simpler way to build, train, and evaluate neural networks. Keras 3 supports TensorFlow, PyTorch, and JAX backends.

Keras: getting started

HUGGING FACE

MODEL HUB

Pretrained models and datasets

Choose to discover, compare, and reuse models and datasets. Read model cards for task fit, evaluation, limitations, and license.

Hugging Face: model cards

HOW THEY FIT TOGETHER Keras runs on TensorFlow or PyTorch; Hugging Face helps you reuse models with them.

FACT-CHECKING AND FURTHER READING

Links checked 10 Nov 2026

1. Google Developers - Problem Framing
2. MIT Open Learning - Neural Network Architectures
3. Harvard / MIT Press - DNN Architectures
4. NVIDIA Developer - Deployment constraints
5. Keras - About Keras 3
6. TensorFlow - Guides
7. Hugging Face - Model Cards

NVIDIA covers deployment; MIT and Harvard/MIT Press offer deeper architecture and evaluation context.

AI disclosure: ChatGPT/Codex supported brainstorming and class preparation. Dr. Juan Orozco reviewed and adapted the material.

DISCUSSION AND REFLECTION

Close with a model-selection conversation

Ask students to explain their reasoning, not guess the “perfect” architecture.

01

Frame the task

In one sentence, what goes in, what comes out, and how will you know the model is good? Name one baseline and one pretrained starting point. How would you compare them fairly?

02

Look beyond the score

If two models have the same accuracy, what else might make you choose one over the other? Consider speed, data needs, and the cost of different errors.

03

Question the evidence

What could make a model look successful during training but perform poorly on new examples? What evidence would help you trust the result?

Quick format: 30 seconds to think, 1 minute in pairs, then invite two students to share.

Remember: clear questions, fair tests, and honest reflection beat searching for a perfect model.